SANKUR KUNDU

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10 fixes merged in 18 days at NVIDIA, Apple, Meta, CMU, Oxford & Hugging Face • work under review by NASA CARA

SUMMARY

ML systems engineer and independent researcher (B.Tech CSE, 2026), sole author of **three manuscripts** and co-founder of Gigzen. Ten fixes merged in 18 days into **NVIDIA's Ising-Decoding** (two), **Apple MLX**, **Meta ExecuTorch** (two), **Carnegie Mellon's XGrammar** (the default structured-output engine of vLLM, SGLang and NVIDIA TensorRT-LLM), **Hugging Face OpenEnv** (two), **Oxford's PyZX** and **JaxMARL**, and I built and submitted a full lab to **Harvard's Machine Learning Systems textbook** (Prof. Vijay Janapa Reddi). I build systems from scratch and benchmark them against the leaders: GPU kernels at **cuBLAS** parity, an LLM engine matching **vLLM** token for token, a collision-risk engine matching **NASA CARA** on all 53 of its test cases, a 494M-parameter LLM that gives **identical bits on an NVIDIA GPU and a CPU**, a gravitational-wave search, built from its own general-relativity waveforms, that finds **62 of 71** LIGO O4a detections blind, and a test showing the Vesuvius Challenge's official ink model finds **79% of real ink** moved into an unread Herculaneum scroll.

HONORS & RECOGNITION

- **Harvard's ML Systems textbook** (Prof. Vijay Janapa Reddi): **built and submitted a full lab**, The Same-Bits Contract, on why temperature-0 LLMs are not deterministic (**PR #2175**, Nov 2026). Built to the book's lab protocol, with its math as a new module of the book's simulator, it passes the book's test suites (**2,658 tests, 0 failures**); its team: "the feature looks great" (discussion).
- **NASA CARA** (Conjunction Assessment Risk Analysis): my independent reproduction of CARA's Monte Carlo results on all 53 of its test cases is **under review by NASA CARA**: its Operations Team Lead referred it to CARA's analysts (20 Oct 2026; emails).
- **Entrepreneur First** (UK): its Talent Investor scouted me from public work alone (15 Oct 2026), citing my cuBLAS-parity compiler and a negative result I chose to publish, "which I rarely see" (email).
- **Upwork Rising Talent** (1 Nov 2026): offered to fewer than 5% of freelancers, for qualifications inside and outside Upwork (email).

OPEN-SOURCE CONTRIBUTIONS: TEN FIXES MERGED IN 18 DAYS, 16 OCT – 2 NOV 2026

NVIDIA ISING-DECODING | NVIDIA's AI decoders for quantum error correction | #98, #99 **Merged 30 Oct & 2 Nov 2026**

- The tests and offline docs looked only for .pt checkpoints, so users of the SafeTensors models NVIDIA publishes on Hugging Face failed them. The tests now find the newest published file by version (v1.0.77 after v1.0.9), and a new test checks that the newest of two exported releases loads bit for bit. NVIDIA's team: "Thanks for the contribution!" After review rounds with maintainer Ivan Basov, he approved and merged it, 30 Oct: "Merged, thank you for the contribution and for working through the review rounds."
- **Second fix,** #99, split out of #98 at his request: the loader ignored the documented PREDECODER_MODEL_CHECKPOINT_FILE for .pt checkpoints; it is now honoured, with a test showing the SafeTensors setting takes precedence, as the README states. After NVIDIA's CI ran it, he approved and merged it, 2 Nov: "Merged, thank you for the follow-up and for adding the precedence test."

APPLE MLX | Apple's ML framework for Apple silicon, **29k★** | PR #4737 **Merged 22 Oct 2026**

- Fixed a data race in the distributed ring backend, the TCP transport that links Macs for multi-machine training: socket threads read their task queues outside the lock. A ThreadSanitizer harness found 2–4 races on every run before the fix and **0** after; MLX's ring tests pass on 8 ranks. Reviewed, approved and merged by MLX maintainer @zcbenz, who also pushed a follow-up commit (22 Oct).

META EXECUTORCH | On-device AI in Instagram & WhatsApp, **5.1k★** | #23677, #23912 **Merged 16 & 28 Oct 2026**

- The 2- and 4-bit quantized embedding kernels of on-device LLMs read memory outside the table on an invalid token id and returned it as an embedding; added the bounds check the 8-bit kernel has, with tests that fail without it (quantized suite: 82 of 82 pass). Approved by Meta maintainers Scott Roy ("LGTM! Thanks for the fix!") and Jacob Stevens; merged 16 Oct.
- **Second fix,** #23912: zero points were checked against the output type, not the scales' type, so valid fp16-scale models aborted, and the .out variants read past their scales (AddressSanitizer: heap-buffer-overflow); after Meta maintainer review, the 2- and 4-bit export kernels also report the scales' dtype. 86/86 kernel tests pass; the new ones fail without the fixes. Merged by Jacob Stevens, 28 Oct.

CMU XGRAMMAR | Default structured-output engine of vLLM, SGLang & NVIDIA TensorRT-LLM | PR #1012 **Merged 29 Oct 2026**

- find_jump_forward_string() raised UnicodeDecodeError whenever the bytes a grammar forces stop partway through a UTF-8 character, as in a JSON-schema enum such as ["é", "è"] or a byte-level vocabulary that splits a character; reported as issue #132, it had been closed by switching jump-forward decoding off. It now returns the longest prefix of complete, valid UTF-8 characters (as strict as Python's decoder), which the grammar still forces, so results stay correct. Approved and merged by maintainer Linzhang Li, 29 Oct.

OXFORD PYZX | ZX-calculus quantum-circuit optimiser, **553★** | PR #563 **Merged 22 Oct 2026**

- Found by checking optimised circuits against the originals that three optimisers put an X rotation on the wrong qubit after rewriting two CNOTs as a SWAP, changing the unitary of **25 of 100** random Clifford circuits (0 after the fix); all 5 automated checks passed. Merged by co-creator John van de Wetering, 22 Oct: "Thanks for the fix! This was a regression from an earlier PR."

JAXMARL | NeurIPS 2024; BOLD Lab: Oxford, UCL & Imperial, **868★** | PR #206 **Merged 26 Oct 2026**

- The rollout manager behind 8 Q-learning baselines (QMIX, VDN, IQL and others) zeroed the first observation of every new episode after an auto-reset, so agents chose their first action blind and trained on it; fixed while keeping zeros for agents that die mid-episode, and two of three new tests fail without the fix. Approved and merged by lead author Alex Rutherford, 26 Oct: "Good find!!"

HUGGING FACE OPENENV | RL environments (Meta-PyTorch, NVIDIA, Microsoft), **2.8k★** | #1441, #1442 **Merged 19 Oct 2026**

- Fixed composite reward rubrics for RL training that ran pre-hooks after scoring their children (a timer read 0.01 ms for 400 ms of work) and scored an async judge as a coroutine; 9 of 13 new tests fail without the fixes. Both merged 19 Oct: "177 rubric tests pass".

In review

- **Google JAX** #41859 (**36.5k★**; Google assigned it to JAX co-author Matthew Johnson, 21 Oct): jax.grad through lax.reduce_window with a custom reduction has failed since 2021; its derivative is now a fold over the window offsets; the **242** new test cases fail on main.
- **vLLM** #60955 (**95.7k★**; labelled a bug for its NVIDIA project): FlashInfer prefill tests compared non-causal attention with a causal reference; against a float64 reference, all 32 cases now pass and all 32 fail on the bug.
- **SGLang** #47305 (**38.4k★**; LMSYS): DeepSeek-V3 tool calls streamed with **empty arguments** when multi-token prediction sent the closing fence and end tokens together; 2 new tests fail without the fix and all 263 parser tests pass.
- **PyTorch** #199875, #199441, #199646: torch.compile now fills torch.empty in deterministic mode as eager does, without filling Inductor's own buffers (why an earlier fix was reverted); re-enabled 8 determinism tests and removed 13 stale skips.
- **MIT Han Lab** nunchaku #957: a lone 4-bit LoRA ignored strengths below 1.
- **ETH Zurich** DaCe #2729: compiled numpy.rint rounded halves away from zero (2.5 → 3; NumPy gives 2); also filed a NaN bug in max/min (#2730). Maintainer: "Seems all good … once green can be merged."

Bug confirmed by the maintainer, fix submitted

- **IBM Relay-BP** #47: IBM's reference decoder answered the same shot differently by batch order; maintainer Tristan Mueller chose my first fix ("thank you for opening this! I think 1. is fine!"), which I submitted (**PR #49**): mismatches **1,970/2,000 → 0**.

MANUSCRIPTS AND RELEASES

- S. Kundu. "Removing the Token Tax from an Existing Model: A Vocabulary Extension for 24 Indian Languages That Keeps Every Other Token." Manuscript, 2026; arXiv (cs.CL) submission in progress (paper).
- S. Kundu. "The Sampler Was Not the Policy: Finding and Removing a Silent Training–Inference Mismatch in a From-Scratch GRPO Stack." Technical report, 2026; arXiv (cs.LG) submission in progress (paper).
- S. Kundu. "Reproducing NASA CARA's Monte Carlo Probability of Collision: An Open Two-Body Implementation, Checked on All 53 of CARA's Test Conjunctions." Manuscript, 2026; arXiv (physics.space-ph) submission in progress (paper).
- Released lipi-Indic 96K (on Hugging Face) and samebit, both Apache-2.0.

ML SYSTEMS: TRAINING, INFERENCE AND RL

ratchet | PyTorch, C++/CUDA: reinforcement-learning post-training (GRPO) from scratch | study page

- Lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** in 100 steps on 2× T4, scoring all 1,319 test problems.
- Cut weight sync **369×** with in-place GPU updates, and made steps **1.9×** faster with partial rollouts and one-step-ahead training.
- Found fp16 weight sync silently erasing most updates, so the sampler ran a policy the trainer never had (gap **5.2 → 0.04** nats).

relay | C++, CUDA, Swift: a disaggregated, fault-tolerant LLM inference cluster | study page

- Wrote flash-attention and flash-decoding kernels that cut 2,048-token prefill **8.7×** (3.7 → 0.43 s) and batch-32 decode **5.4×** on a T4.
- Streamed the KV cache layer by layer, hiding **99–100%** of the transfer: a 46 MB hand-off went from 21.6 to **0.08 ms**.
- Kept output identical through failures: watchdogs reroute requests from killed, frozen or stalled workers (**384/384 through 60 kills**; none lost in 300 fault-injection seeds).

samebit | C, CUDA, PyTorch, MLX: LLM inference in integers, the same bits on every chip

- Ran **Qwen2.5-0.5B** (494M parameters, 24 layers, grouped-query attention) **entirely in integers** (int8 weights, int16 activations, int32 residual stream, int64 sums): WikiText-2 perplexity **23.16 vs 22.99** for float32 (**+0.75%**); 93.25% of top-1 tokens unchanged.
- Showed **bit-identity on an NVIDIA Tesla T4** (public Kaggle run): it reproduces the x86 CPU's logit digests on real text, with each dot product split over 1–64 blocks and summed by atomicAdd in a new order every run; float16 shifts logits by up to **0.25**.
- Made **six independent backends** (C99/OpenMP, NumPy, PyTorch, **Apple MLX**, **CUDA**, reference) give **one digest over all 153 intermediate tensors** on x86-64, ARM64 and the T4, checked in CI on every push.
- Built **The Same-Bits Contract**, a full lab **submitted to Harvard's Machine Learning Systems textbook**: up to **7 of 64** users get different text at temperature 0 from batching alone, fixed two ways.

switchyard | CUDA, C++: mixture-of-experts serving with expert parallelism | study page

- Wrote routing, sorting and grouped expert GEMM/GEMV kernels on tensor cores and peer-to-peer dispatch: OLMoE-1B-7B decodes **2.2× faster than Hugging Face**, bit-identical to one GPU, with dispatch **1.25× faster than NCCL**.
- Showed on real text that dropping tokens at full experts raises perplexity from 12.9 to 20.2 even at capacity factor 2, so the kernels never drop one; found and fixed two cross-GPU stream races.

crucible | Python on Linux: sandboxed environments for RL on coding agents | study page

- Sandboxes from namespaces, seccomp-BPF and limits fork in **4.5 ms** (vs 138 ms); the grader passes **0 of 9** known reward hacks.
- Trained with GRPO on ratchet, every reward from a sandbox: MBPP pass@1 **34.9% → 38.6%** (3 seeds; paired test, p = 0.013).

tandem | C++, CUDA: distributed training without torch.distributed or NCCL | study page

- Built DDP, ZeRO-1/2/3, GPipe and 1F1B on a custom ring all-reduce that trains **bit-identically to PyTorch DDP/FSDP on NCCL**; ZeRO-3 cuts per-GPU memory 17% on an 86M-parameter GPT.

GPU KERNELS AND COMPILERS

tessel | Rust, zero dependencies: a tile language and compiler for GPU and TPU kernels | study page

- Wrote a Triton-style compiler: typed tile IR, linear layouts, tensor-core CUDA (ldmatrix, mma.sync, swizzled shared memory, software pipelining), an auto-tuner, and Metal and TPU (Pallas) backends.
- Matched **cuBLAS** on GEMM (**1.00–1.04×**) and beat PyTorch SDPA on causal attention (**1.3–1.9×**), torch.compile on RMSNorm (1.27×) and Triton on softmax (1.07×) on a T4, checked against fp64.
- Built an LLM engine on these kernels alone (paged KV cache, continuous batching, CUDA graphs) that matches **vLLM token for token at 1.05× its throughput** on TinyLlama-1.1B.

kiln | Rust, zero dependencies: an ML compiler from ONNX to SIMD C and CUDA | study page

- Gave the lowest latency on all three benchmark models on 4 CPU cores: BERT in **117 ms** vs 149 for torch.compile, 153 for ONNX Runtime and 189 for JAX/XLA; on a T4, SmolLM2 **1.65×** faster than torch.compile.

SCIENCE: GRAVITATIONAL WAVES, THE HERCULANEUM SCROLLS, SPACE SAFETY AND QUANTUM ERROR CORRECTION

chirp | Python, PyTorch (CUDA): an independent search for gravitational waves in public LIGO data

- Wrote the 3.5PN TaylorF2 waveform from general relativity; it matches LIGO's LALSimulation **within a billionth of a radian**.
- Searched every two-detector event of the LIGO–Virgo–KAGRA O4a catalog (GWTC-4.0) with a 3,886-template bank on one T4 in **17 minutes**: **all 71** with complete data re-found at a median **95%** of the catalog's network SNR, 67 within 50 ms (public run).
- Added gating and the Allen chi-squared veto with reweighted SNR, as LIGO's PyCBC searches use: searched blind over 188 s of data per event, the loudest trigger is the real event for **62 of 71** (48 without the veto), at a cost of 2% of the events' own SNR (public run).

ink-sensitivity | Python, PyTorch: calibrating ink detection for the unread Herculaneum scrolls (Vesuvius Challenge)

- Transplanted real ink from PHerc. 139, located by its published 2.4 µm ink map, into the CT volume of **PHerc. 800**, an unread scroll eligible for the challenge's prizes; the control moves bare papyrus through the same stroke shapes, so the two differ only in the ink.
- The official 9 µm model finds **79%** of the real-ink windows at a 5% false-positive rate (**AUC 0.94**; 168 ink vs 168 controls, no untouched window flagged), and transplanted ink scores like the same text left in place.
- Found that reading the volume in reversed depth order leaves the model nearly blind (**0 of 48** real-text windows), so averaging both directions dilutes it; a pure brightness profile of ink, even at 4×, is barely seen.

kessler | C++/CUDA, Python: an independent auditor of space traffic | live globe

- Matched **NASA CARA's** Monte Carlo collision probability on **all 53** of its test conjunctions (chi-squared p = 0.67), including the 28 where the standard 2D method is wrong, **15–19× faster** per core than NASA's library.
- Screens all **531 million pairs** of 32,593 tracked objects a day ahead in **28 s** on a T4, matching CelesTrak SOCRATES to 0.24 m (median); SGP4 runs 125.7M orbit states/s in fp64.

syndrome | C++, CUDA: an open GPU decoder for IBM's quantum LDPC codes | study page

- Matched IBM's Relay-BP reference **bit for bit on 256/256 shots** of the [[144,12,12]] gross code (391,320 edges).
- Decoded a single shot **28.7× faster than IBM's decoder** (median 37.5 ms vs 1.08 s) and a batch 4.0× faster than IBM's on all CPU threads, after reproducing IBM's published error rate (41 vs 44 in 10,000).

parity | Python, Stim, PyTorch: an independent benchmark of quantum error-correction decoders

- Ran NVIDIA's two published Ising pre-decoders through NVIDIA's own pipeline (d = 5–13, 524,288 shots each), matching its logged error rates to every printed digit, against PyMatching on identical samples.
- Reproduced PyMatching 2 / Sparse Blossom (Higgott & Gidney): scaling exponent 1.38 (paper: 1.32), thresholds at 0.61–0.80% for d = 3–11, and 0 logical errors in 1M shots per basis at d = 17.

LANGUAGE AI

lipi | PyTorch: cutting the token tax on Indian languages | study page

- Measured 24 Indian languages on 15 tokenizers (Odia: 4.99× English on GPT-4o) and built lipi-Indic (on Hugging Face), which needs fewer tokens than **GPT-4o's and Gemini's in 23 of 24** (1.30× vs 1.92×, 1.58×).
- Kept English, code and other scripts on their Qwen 2.5 ids; the 96K new merges fire only on Indian scripts (Santali 13.7× → 1.52×).
- Extended Qwen2.5-0.5B and continued pre-training on 2× T4: Odia takes **8.6×** fewer tokens and, in equal GPU time, is predicted **20% better**, with Odia→English chrF++ 16.0 → 24.8.

DISTRIBUTED SYSTEMS AND VERIFICATION

quorumdb | Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible | study page

- Built LSM storage, self-rebalancing multi-Raft, serializable transactions (Percolator) and a SQL planner: **1,500 tpmC** (TPC-C-like).
- Found **6 real bugs** with deterministic simulation (**10,000 Raft clusters, 610K crashes**, histories machine-checked) and a TLA+ model (337K states), which also catch **all 15 planted bugs**.

faultline | Rust, zero dependencies: Jepsen-style fault injection for etcd

- Checked 1.26M operations under 288 faults across 4 etcd release lines for linearizability, and traced lost updates under etcd locks to a paused leader applying a timed-out write 6 s late; fenced writes lost none.

EXPERIENCE

GIGZEN | Co-founder & Founding Engineer | Bhubaneswar, India **Apr 2026 – Present**

Early-stage startup (company name approved by India's MCA; incorporation in progress) | Sole engineer across every product

- Built **Waggle**, a delivery platform of three Android apps (riders, shops, customers) from one TypeScript/React/Capacitor codebase on Supabase PostgreSQL: **53 tables, 90 RLS policies, 154 SQL functions**.
- Moved first-accept-wins dispatch, UPI reconciliation and GST billing into the database, where no client can bypass them, and tested its access rules with **150 automated attacks**, all blocked.
- Released **Populace**, an open-source tool that fills an app with simulated users acting through its real API (npm, **1,600+ downloads**); its first run found **5 defects** that manual testing had missed.

TECHNICAL SKILLS

GPU and compilers: CUDA (tensor cores, mma.sync, ldmatrix, graphs, IPC), NVRTC, Triton-style tile languages, Metal, Pallas, SIMD
ML systems: GRPO, DDP, ZeRO-1/2/3, FSDP, 1F1B, MoE expert parallelism, paged KV cache, int8 quantization, deterministic inference
Distributed systems: Raft, multi-Raft, MVCC, Percolator, LSM trees, deterministic simulation, TLA+, fault injection
Languages and platforms: C++20, CUDA, C, Rust, Python, Swift, TypeScript, SQL; PyTorch, JAX, MLX, PostgreSQL, Docker, Kubernetes

EDUCATION

IEM KOLKATA | B.Tech, Computer Science and Engineering (AI & ML) | Kolkata, India **Sep 2022 – Apr 2026**